

B. PHARM.
(SEM VIII) THEORY EXAMINATION 2022-23
BIOSTATISTICS AND RESEARCH METHODOLOGY

Time: 3 Hours**Total Marks: 75****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.****2 x 10 = 20**

- For a binomial distribution with $p = 0.25$ and $n = 10$, find mean and standard deviation.
- Calculate arithmetic mean for the following data: 78, 52, 82, 38, 69, 71
- Calculate median for the following data: 10, 6, 2, 3, 8, 15, 12, 5
- Find mode value for the given data: 2, 2, 3, 4, 7, 7, 7, 7, 9, 10, 12, 12
- What are the Type I and Type II errors?
- Write the definition of probability.
- What is source- based plagiarism?
- What do you mean by randomized controlled design?
- What do you mean by negative correlation?
- What do you understand by cohort design?

SECTION B**2. Attempt any two parts of the following:****10 x 2 = 20**

- Height of fathers and sons in inches are given below:

Height of father	65	66	67	67	68	69	70	71
Height of son	66	68	65	69	74	73	72	70

Find the two lines of regression and calculate the estimated average height of son when the height of father is 68.5 inches.

- Explain and design different phases of clinical trial.
- Differentiate between case-control and cohort studies in detail.

SECTION C**3. Attempt any five parts of the following:****7 x 5 = 35**

- Calculate standard deviation for the following data:

Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency	5	8	15	16	6

- Calculate the coefficient of correlation from the data given below:

x	2	3	4	5	6	7	8
y	4	6	8	9	12	15	16

- The chances of catching cold by workers working an ice factory during winter are 25%. What is the probability (using binomial distribution) that out of 5 workers 4 or more will catch cold?
- Explain Blocking and confounding system for Two-level factorials.
- What is Report writing and presentation of data? Explain the steps of report writing.
- Explain about the objective of plagiarism and the types of plagiarism.
- What do you understand by Response Surface methodology? Explain it.